

Weekly Research Progress Report

Baseline Evaluation of EfficientNetV2B0 for Multi-Class Plant Disease Classification

Status: Baseline experiment completed; advanced experiments are in progress.

1. Introduction

Plant diseases can significantly affect agricultural productivity. Early and accurate identification of plant diseases from leaf images can assist farmers and agricultural professionals in taking timely corrective measures.

Deep learning, particularly Convolutional Neural Networks (CNNs), has shown strong performance in image-based plant disease classification. In this research work, the initial objective is to establish a strong CNN-based baseline using a modern pretrained architecture. The baseline will later be extended toward robustness, efficiency, explainability, and generalization under real-world conditions.

2. Research Objective

The objective of this stage is to develop and evaluate a modern CNN-based baseline model for multi-class plant disease classification using transfer learning and establish reference performance for subsequent research experiments.

The baseline will provide a point of comparison for advanced CNN architectures, attention mechanisms, image degradation and robustness, cross-dataset validation, explainable AI, and computational efficiency.

3. Dataset

The experiments were conducted using a **15-class subset of the PlantVillage dataset**. The dataset contains **20,638 images** across 15 classes. Images were resized to 224×224 pixels.

The data was divided approximately into 70% training, 15% validation, and 15% testing.

No.	Class
1	Pepper bell Bacterial spot
2	Pepper bell healthy
3	Potato Early blight
4	Potato Late blight
5	Potato healthy
6	Tomato Bacterial spot
7	Tomato Early blight
8	Tomato Late blight
9	Tomato Leaf Mold
10	Tomato Septoria leaf spot
11	Tomato Spider mites
12	Tomato Target Spot
13	Tomato Yellow Leaf Curl Virus
14	Tomato mosaic virus

No.	Class
15	Tomato healthy

4. Methodology

The experimental pipeline was: **PlantVillage Dataset** → **Data Splitting** → **Data Augmentation** → **EfficientNetV2B0** → **Feature Extraction** → **Global Average Pooling** → **Dropout** → **15-Class Softmax Classifier** → **Evaluation**.

Data preprocessing: All images were resized to $224 \times 224 \times 3$. A batch size of 32 was used. Training data augmentation consisted of random horizontal flipping and random rotation.

5. CNN Architecture

EfficientNetV2B0 was selected as the baseline architecture. The model was initialized with ImageNet pretrained weights and the original classification head was removed. The architecture consisted of an input layer, data augmentation, EfficientNetV2B0 pretrained backbone, Global Average Pooling, Dropout (0.3), and a Dense softmax classification layer for 15 classes. The pretrained backbone was initially frozen.

6. Training Configuration

Parameter	Configuration
Framework	TensorFlow / Keras
TensorFlow version	2.20.0
CNN architecture	EfficientNetV2B0
Pretrained weights	ImageNet
Input size	224×224
Batch size	32
Number of classes	15
Optimizer	Adam
Learning rate	0.001
Loss function	Sparse Categorical Crossentropy
Data augmentation	Random Flip + Random Rotation
Dropout	0.3
Maximum epochs	10
Early stopping	Enabled; patience = 3

7. Training Results

The model showed progressive improvement during training. The following recorded epochs illustrate the training trend:

Epoch	Training Accuracy	Validation Accuracy	Training Loss	Validation Loss
1	71.83%	87.46%	1.0164	0.5100
2	85.53%	90.86%	0.5043	0.3524
3	88.00%	91.51%	0.4011	0.2945

8. Test Results

The final evaluation was performed on 3,104 test images. The direct model evaluation produced a **test loss of 0.1771** and **test accuracy of 94.81%**.

Metric	Score
Accuracy	94.75%
Macro Precision	94.54%
Macro Recall	94.87%
Macro F1-score	94.58%

9. Classification Analysis

A classification report was generated for all 15 disease and healthy categories. The model demonstrated strong class-level performance. For example, Pepper bell Bacterial spot achieved 98.09% precision, 100.00% recall, and 99.04% F1-score; Pepper bell healthy achieved 99.57% precision, 99.15% recall, and 99.36% F1-score; Potato Early blight achieved 96.62% precision, 99.31% recall, and 97.95% F1-score.

A confusion matrix was also generated to investigate class-level prediction behavior and identify categories with comparatively more misclassifications.

10. Interpretation of Results

The baseline experiment demonstrates that EfficientNetV2B0 can achieve strong classification performance on this 15-class PlantVillage subset. The approximately 94.8% test accuracy indicates that transfer learning using a pretrained CNN provides an effective starting point for plant disease recognition.

However, high performance on PlantVillage alone does not establish that the model will perform equally well on real-world agricultural images. Real-world photographs can contain different illumination, blur, compression, background variation, camera quality, leaf orientation, and lower resolution.

11. Current Research Gap

A model that performs well on a controlled dataset may not necessarily remain reliable when image quality and visual conditions change. Therefore, the next stages will investigate whether a lightweight CNN can maintain reliable disease classification under realistic image degradation while remaining computationally efficient and interpretable.

12. Future Work

Modern CNN comparison: ConvNeXt-Tiny, other EfficientNetV2 variants, and a proposed lightweight CNN will be evaluated.

Attention mechanism: CBAM or a related attention mechanism will be investigated.

Robustness evaluation: Gaussian noise, blur, brightness/contrast variation, JPEG compression, and reduced resolution will be tested.

Cross-dataset validation: External validation using a dataset such as PlantDoc will be considered.

Explainability: Grad-CAM will be used to visualize disease-relevant regions.

Ablation study: The contributions of augmentation, attention, and other components will be evaluated.

Computational analysis: Parameter count, model size, inference time, and computational complexity will be compared.

13. Conclusion

In this week's research work, a CNN-based baseline for plant disease classification was successfully implemented using EfficientNetV2B0 with ImageNet transfer learning. The experiment used a 15-class PlantVillage subset containing 20,638 images. The model achieved approximately 94.81% test accuracy, with a macro F1-score of approximately 94.58%.

These results establish a strong baseline for subsequent research stages. The next stage will focus on modern CNN comparison, attention mechanisms, robustness testing, explainability, and external validation.